

# **LAGUNA STATE POLYTECHNIC UNIVERSITY**

## **BIOS vs UEFI**

Technical Comparison and Modern Deployment Rationale

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## Comparison Table

| Criterion       | Legacy BIOS                                                                   | UEFI                                                                   |
|-----------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Definition      | Legacy firmware that initializes hardware and transfers control to boot code. | Modern standardized firmware with a boot manager and EFI applications. |
| Boot process    | Reads code from the MBR/boot sector.                                          | Uses NVRAM entries to load an EFI application from the ESP.            |
| Disk support    | Usually MBR; traditional limit about 2 TiB with 512-byte sectors.             | Usually GPT; supports larger disks and more partitions.                |
| Partition style | MBR; typically four primary entries.                                          | GPT; more flexible partitioning.                                       |
| Security        | No standardized Secure Boot chain.                                            | Supports Secure Boot when configured.                                  |
| Speed           | Legacy initialization may be slower.                                          | Modern initialization can support faster startup.                      |
| Advantages      | Older-system compatibility and simple legacy behavior.                        | GPT, modern device support, flexible boot management, Secure Boot.     |
| Disadvantages   | Legacy disk/boot constraints and limited pre-boot security.                   | Legacy compatibility can require alternate settings.                   |
| Modern usage    | Older or specialized compatibility scenarios.                                 | Default model on contemporary PCs and servers.                         |

## Why UEFI Has Largely Replaced BIOS

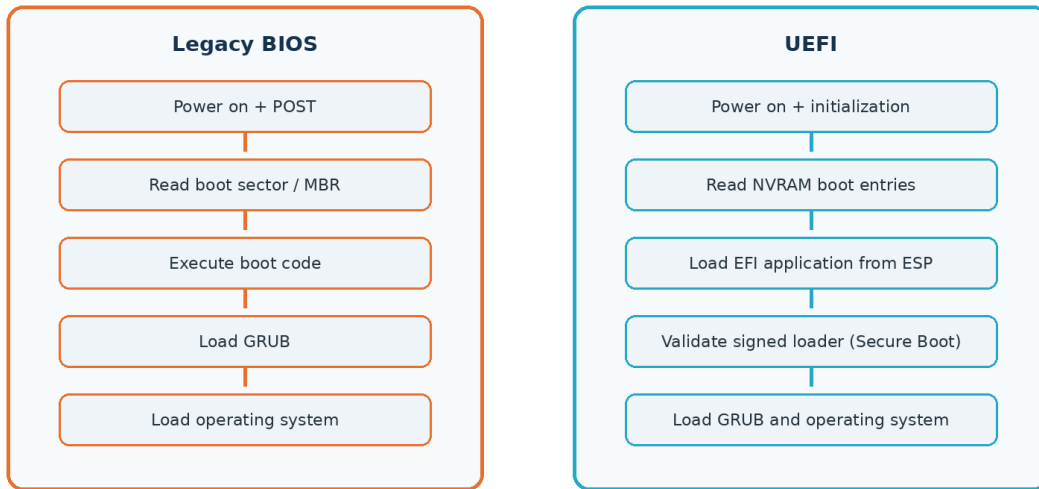
UEFI has largely replaced legacy BIOS because modern systems need a more capable and standardized method to initialize hardware and locate an operating system. Traditional BIOS normally transfers control through boot code in the master boot record, a design with historical constraints in disk layout, boot management, extensibility, and pre-boot security.

UEFI defines firmware services and a firmware boot manager. Boot entries are stored in nonvolatile variables, and each entry can identify an EFI application on a device, normally an operating-system loader on the EFI System Partition. UEFI systems commonly use GPT, removing the familiar MBR partition-count restriction and supporting storage beyond the traditional MBR addressing limit.

UEFI Secure Boot can verify signatures on boot components against keys trusted by the platform. Properly configured, this helps prevent unauthorized or modified pre-boot software from executing. It complements rather than replaces patching, encryption, access control, and monitoring.

Legacy BIOS remains relevant for older hardware or operating systems. Administrators must match firmware mode, partition style, and boot loader. Changing modes without preparing the disk and boot files can make a system unbootable. For new supported servers, UEFI with GPT is generally the appropriate baseline, with Secure Boot enabled when supported by policy and software.

## Firmware Boot Paths: Legacy BIOS and UEFI



Firmware boot paths: Legacy BIOS and UEFI

### Key Takeaway

Select firmware mode before installation, keep it consistent with the disk layout and boot loader, and document the setting. Prefer UEFI and GPT for new supported servers unless a verified compatibility requirement dictates otherwise.